Concordia UNIVERSITY

Ordinary Differential Equations

Higher-Order Differential Equations

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ENGR 213

Lecture 11 (October 21, 2026)

Today's Lecture

- Linear Equations with Constant Coefficients
  - Homogeneous linear DEs
  - Auxiliary equation
  - Higher-order Equations

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Section 3.3**

- **Homogeneous linear DEs**

  - A homogeneous linear differential equation with constant coefficients:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

where the coefficients a_i, i = 0, 1, …, n are real constants and $a_n \neq 0$.

All solutions of these higher-order DEs are either exponential functions or are constructed out of exponential functions.

  - For a homogeneous linear second-order equation:

$$ay'' + by' + cy = 0$$

- **Auxiliary Equation**

  - For a homogeneous linear second-order equation:

$$ay'' + by' + cy = 0$$

  the auxiliary equation is $\boldsymbol{am^2 + bm + c = 0}$

  - The two roots of $\boldsymbol{am^2 + bm + c = 0}$ are:

$$m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

  - There will be three forms of the general solution corresponding to the three cases:

    - m_1 and m_2 are real and distinct ($b^2 - 4ac > 0$)
    - m_1 and m_2 are real and equal ($b^2 - 4ac = 0$)
    - m_1 and m_2 are conjugate complex numbers ($b^2 - 4ac < 0$)

- **Case I:** m_1 and m_2 are real and distinct $\quad y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

- **Case II:** m_1 and m_2 are real and equal $\quad y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$

- **Case III:** m_1 and m_2 are conjugate complex numbers

$$m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \longrightarrow m_1 = \alpha + i\beta \qquad m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \longrightarrow m_2 = \alpha - i\beta$$

$$y = C_1 e^{(\alpha + i\beta)x} + C_2 e^{(\alpha - i\beta)x}$$

Euler's formula: $e^{i\theta} = \cos\theta + i \sin\theta$ where θ is any real number.

$$y = e^{\alpha x}(c_1 \cos\beta\, x + c_2 \sin\beta\, x)$$


QUESTIONS

EXAMPLE 1

Solve $2y'' - 5y' - 3y = 0$

EXAMPLE 2

Solve $y'' - 10y' + 25y = 0$

EXAMPLE 3

Solve $y'' + 4y' + 7y = 0$

EXAMPLE 4

Solve the IVP $4y'' + 4y' + 17y = 0, \quad y(0)=-1, \quad y'(0)=2.$

- **Higher-Order Equations**

  - In general, to solve an nth-order differential equation:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

where the a_i, i = 0, 1, …, n are real constants and $a_n \neq 0$.

we must solve an nth-degree polynomial equation:

$$a_n m^n + a_{n-1} m^{n-1} + \cdots + a_2 m^2 + a_1 m + a_0 = 0$$

  - If all the roots of auxiliary equation are real and distinct, then the general solution is:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}$$

  - When m_1 is a root of multiplicity k of an nth-degree auxiliary equation (that is, k roots are equal to m_1), the general solution must contain:

$$c_1 e^{m_1 x} + c_2 x e^{m_1 x} + c_3 x^2 e^{m_1 x} + \cdots + c_k x^{k-1} e^{m_1 x}$$


QUESTIONS

EXAMPLE 5

Solve $y''' + 3y'' - 4y = 0$

EXAMPLE 6

Solve $y'''' + 2y'' + y = 0$

EXAMPLE 6

Solve $3y''' + 5y'' + 10y' - 4y = 0$